

Random variables

$X: \Omega \rightarrow \mathbb{R}$
random variable is a function from sample space to real line.

Ex1 $\Omega = \{H, T\}$ $X(\omega) = \begin{cases} 1 & \text{if } \omega = H \\ 0 & \text{if } \omega = T \end{cases}$

Ex2 $\Omega = \{HHH, HHT, THH, \dots, TTT\}$

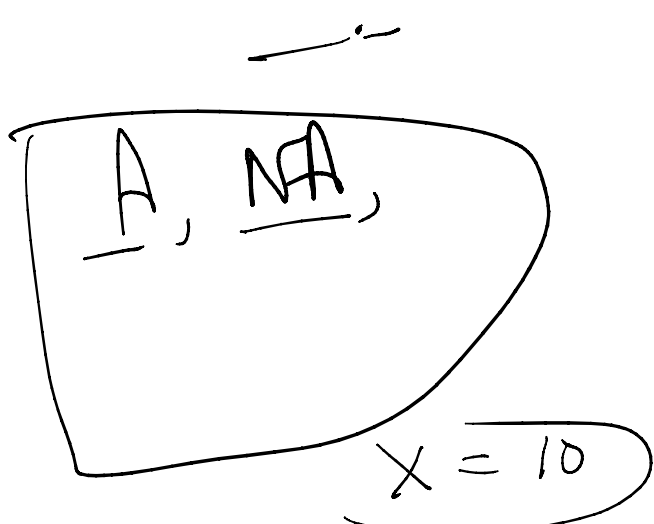
the number of heads $X(\omega) = \begin{cases} 0 & \omega = TTT \\ 1 & \omega = TTH, THT, HTT \\ 2 & \omega = HHT, HTH, THTT \\ 3 & \omega = HHH \end{cases}$

$$P(X=3) = \frac{1}{8}$$

$$P(X=2) = \frac{3}{8}$$

$$P(X=1) = \frac{3}{8}$$

$$P(X=0) = \frac{1}{8}$$



$$P(X=3) + P(X=2) + P(X=1) + P(X=0) = 1.$$

X	$P(X=x)$
0	$\frac{1}{8}$
1	$\frac{3}{8}$
2	$\frac{3}{8}$
3	$\frac{1}{8}$

probability distribution.

discrete random variable can take only some isolated values

$P(X=x) = p(x) \rightarrow$ probability mass function (pmf).
two conditions on pmf
(i) $p(x) \geq 0$
(ii) $\sum_x p(x) = 1$

probability distribution function:

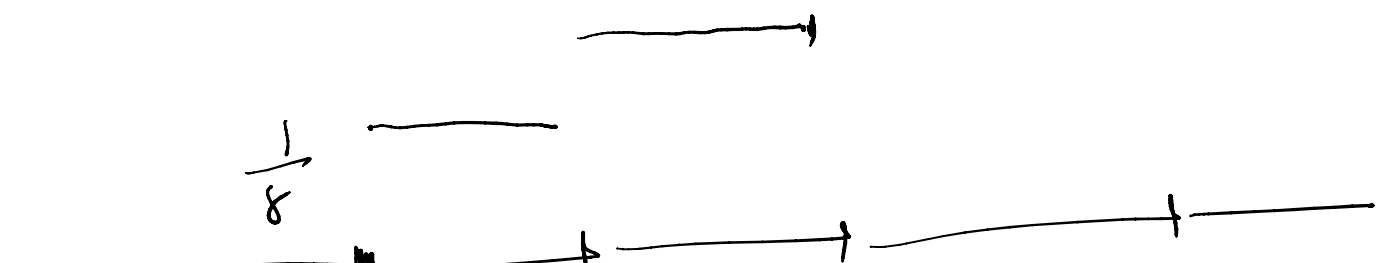
$$F(x) = \sum_{x_i \leq x} P(x_i)$$

In Ex2

$$F(2.5) = P(X=0) + P(X=1) + P(X=2)$$

$$F(0) = P(X=0) = \frac{1}{8}$$

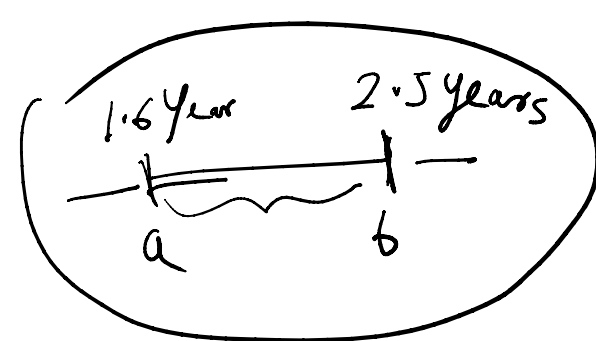
$$F(-1) = 0$$





Continuous random variable can take any value within a certain range.

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$



$f(x)$: probability density function (pdf)

two conditions

- (i) $f(x) \geq 0$
- (ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

A function is said to be pdf if (i) and (ii) are true for that

function.

$F(x)$ - probability distribution function.

$$F(x) = P(X \leq x) = P(-\infty < X \leq x) = \int_{-\infty}^x f(u) du$$

(both for discrete and continuous rv)

- (i) $F(x)$ is increasing
 $x_1 < x_2$
 $F(x_1) < F(x_2)$
- (ii) $F(-\infty) = 0$
- (iii) $F(\infty) = 1$

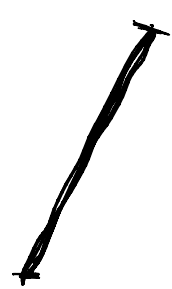
Continuous case

$$f(x) = \frac{dF(x)}{dx}$$

$$P(X = x) = 0$$

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

$$\begin{aligned} &= P(a < X < b) \\ &= P(a \leq X \leq b) \\ &= P(a < X < b) \end{aligned}$$



$$\frac{4 \text{ cm}}{4.001 \text{ cm}}$$

$$3.9998 \text{ cm}$$

$$(4-\epsilon, 4+\epsilon)$$

Ex.

X	$P(X=x)$
1	k
2	$2k$
3	$2k$
4	$3k$
5	k^2
6	$2k^2$
7	$7k^2 + k$

$$F(6) = P(X=1) + P(X=2) + P(X=3) + P(X=4) + P(X=5) + P(X=6)$$

$$= k + 2k + 2k + 3k + k^2 + 2k^2$$

- (i) find k . (ii) $F(6) = ?$ 0.83
- 1/10 0.83

Ex.

$$f(x) = \begin{cases} 2x & 0 < x \leq 1 \\ 4-2x & 1 < x \leq 2 \\ 0 & \text{o.w.} \end{cases}$$

Is $f(x)$ a pdf?

- Sol (i) $f(x) \geq 0$ ✓
- (ii) $\int_{-\infty}^{\infty} f(x) dx = 0 + \int_0^1 2x dx + \int_1^2 (4-2x) dx = 0 + 1 + 1 = 2 > 1$

not pdf.

X is cont rv, pdf $f(x) = \begin{cases} \frac{1}{4} & -2 \leq x \leq 2 \\ 0 & \text{o.w.} \end{cases}$

Ex.

$$P(2X+3 > 5)$$

$$= P(X > 1) = \int_1^{\infty} f(x) dx = \int_1^2 \frac{1}{4} dx = \frac{1}{4}$$

Ex.

$$F(x) = 1 - e^{-\lambda x^\alpha}, \quad x > 0$$

$$f(x) = \frac{dF(x)}{dx} = \lambda \alpha x^{\alpha-1} e^{-\lambda x^\alpha}$$

Ex.

$$F(x) = 1 - e^{-\lambda x}$$

$$P(1 \leq X \leq 3) = ?$$

$$P(1 \leq X \leq 3) = \int_1^3 f(x) dx$$

$$= F(3) - F(1)$$

$$= (1 - e^{-\lambda 3}) - (1 - e^{-\lambda})$$

$$= e^{-\lambda} - e^{-3\lambda}$$

$$P(a \leq X \leq b) = F(b) - F(a)$$

check